



Pharm*Achieve*

OTITIS EXTERNA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ADR Adverse Drug Reaction

BID Twice Daily

C/I Contraindications

DEET N,N-diethyl-3-meta-toluamide

EAC External Auditory Canal

HCP Healthcare Professional

N/V/D Nausea/Vomiting/Diarrhea

OE Otitis Externa

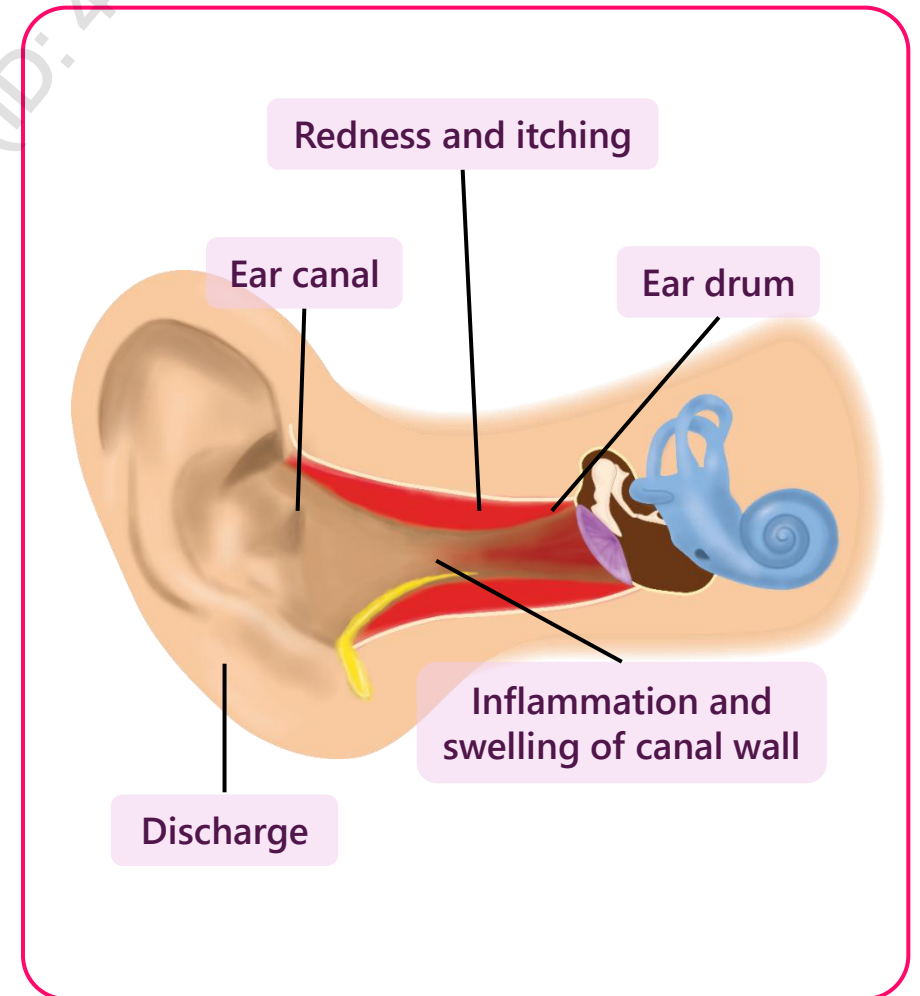
OTC Over-the-Counter

QID Four Times Daily

TID Three Times Daily

PATHOPHYSIOLOGY

- Also known as swimmer's ear
- Inflammation of external ear/auditory canal (EAC); which is the tube between the outer ear and ear drum
- EAC is warm, dark and moist, thus enabling bacteria or fungi to proliferate, particularly if there is already ear trauma
- Occurs more commonly in the summer months because swimming prolongs ear dampness, thus causing skin to shed, which allows organisms to enter and infections to occur
- Types of EAC: **acute diffuse (our focus)**, acute localized, chronic, eczematous or necrotizing
- Etiology for acute diffuse OE is commonly bacterial: *Pseudomonas aeruginosa* (20-60%) and *Staphylococcus aureus* (10-70%) but can be polymicrobial



RISK FACTORS

- Very little cerumen (antibacterial activity by physically protecting canal and maintaining a low pH)
- A lot of cerumen (maceration and occlusion), moisture (e.g. bathing, water sports, sweating, swimming, increased humidity) results in skin maceration and elevated pH
- EAC trauma (e.g. cotton-tip swabs, foreign objects, fingernails) causing abrasion allowing inoculation of organisms
- Chronic skin conditions
- Hearing aids
- Narrow and hairy ear canals

PATIENT ASSESSMENT ACUTE DIFFUSE OTITIS EXTERNA

- Generally, **bacterial acute otitis** externa causes **ear pain/otalgia** (may worsen when chewing food or manipulating the ear), **ear discomfort, otorrhea, pruritus, tenderness**
 - Bacterial otitis externa symptoms are more intense than those seen with fungal OE
- **Fungal otitis externa** rarely occurs
 - May be **asymptomatic** or may cause **pruritus** and **fullness** in the ear
 - Occurs in patients who are **immunocompromised** or in those who have been treated for bacterial otitis externa with antibiotics for prolonged periods of time (prolonged antibiotic treatment alters the EAC bacterial flora)
 - EAC may contain black, grey, bluish-green, yellow, or white fungal elements and debris
 - Common infecting pathogens *Candida albicans* and *Aspergillus* species
- **Unilateral (90% frequency), acute onset (within 48 hours) of otalgia (70% frequency), itching (60% frequency), fullness (frequency 22%), otorrhea, and hearing loss (32% frequency)**
- **Commonly occurs in children ages 7-12 years and frequency declines after age 50**

OTHER TYPES OF OTITIS EXTERNA

Acute localized OE (furunculosis)

- Type of OE where patients present with an infected hair follicle ("boil") in the ear canal
- Common infecting pathogen is *S. aureus*
- Symptoms: edema, erythema, itching, pain (subsides when boil bursts), possible abscess

Eczematous OE

- Related to many dermatologic conditions e.g. psoriasis, lupus, infantile eczema, atopic, seborrheic, or contact dermatitis
- Symptoms: ranging from mild redness and scaling (related to atopic dermatitis) to psoriasis; most common symptom is pruritis and may be accompanied by crusting, oozing, and secondary infections, common to have symptoms on other parts of the body as well

Chronic OE

- OE lasting >3 months or >4 episodes of OE/year
- Thickening of EAC skin due to low-grade infection and swelling
- Painless but continuous pruritus, discomfort, dry flaking skin, lack of cerumen
- Causes are commonly non-bacterial e.g. allergic contact dermatitis

Necrotizing (malignant) OE

- Infection extending to the temporal or mastoid bone most seen in immunocompromised patients e.g. patients with diabetes, >65 years of age
- Common infecting pathogen is *S. aureus* (90%)

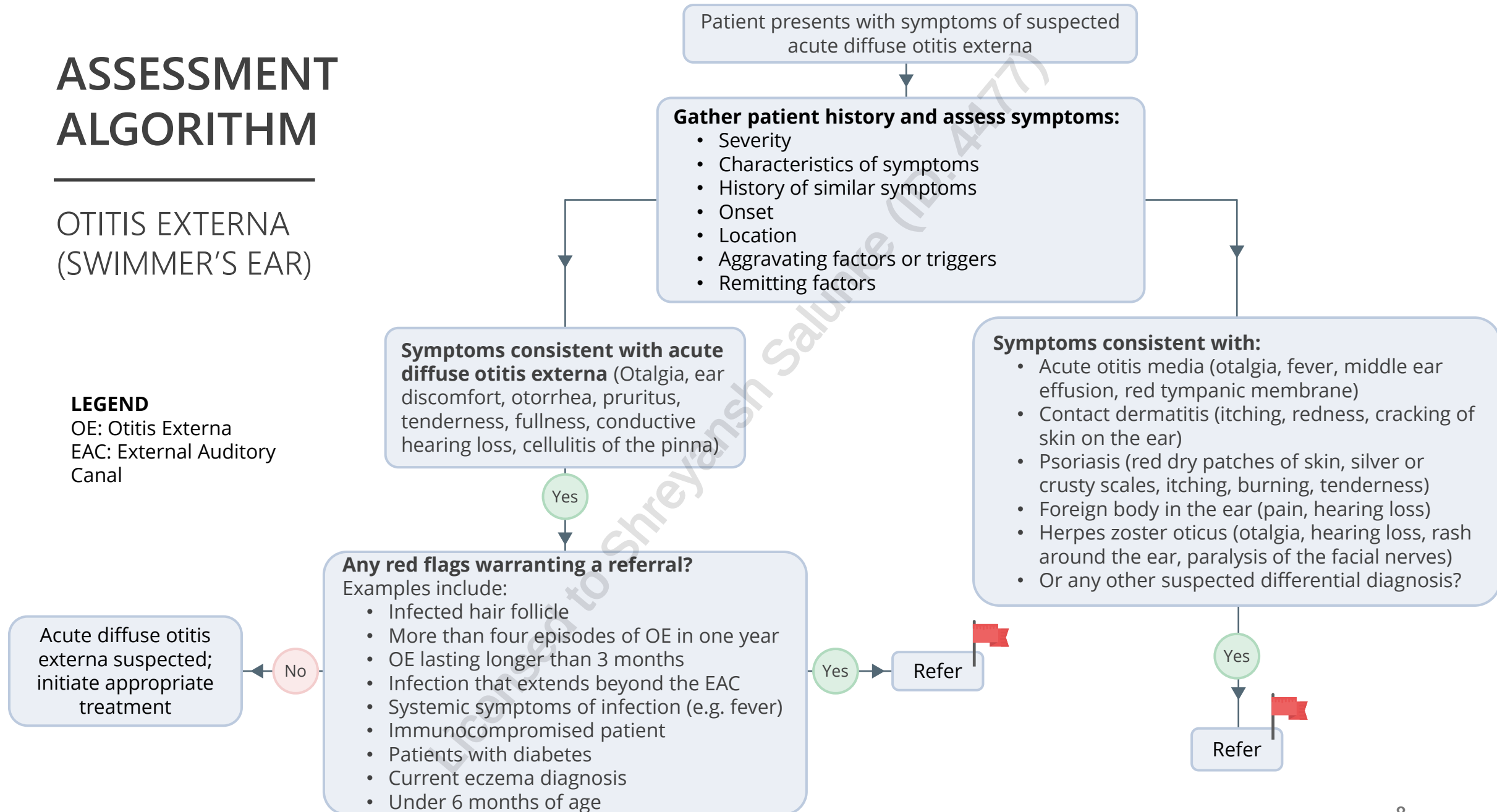
ASSESSMENT ALGORITHM

OTITIS EXTERNA (SWIMMER'S EAR)

LEGEND

OE: Otitis Externa

EAC: External Auditory
Canal



GOALS OF THERAPY

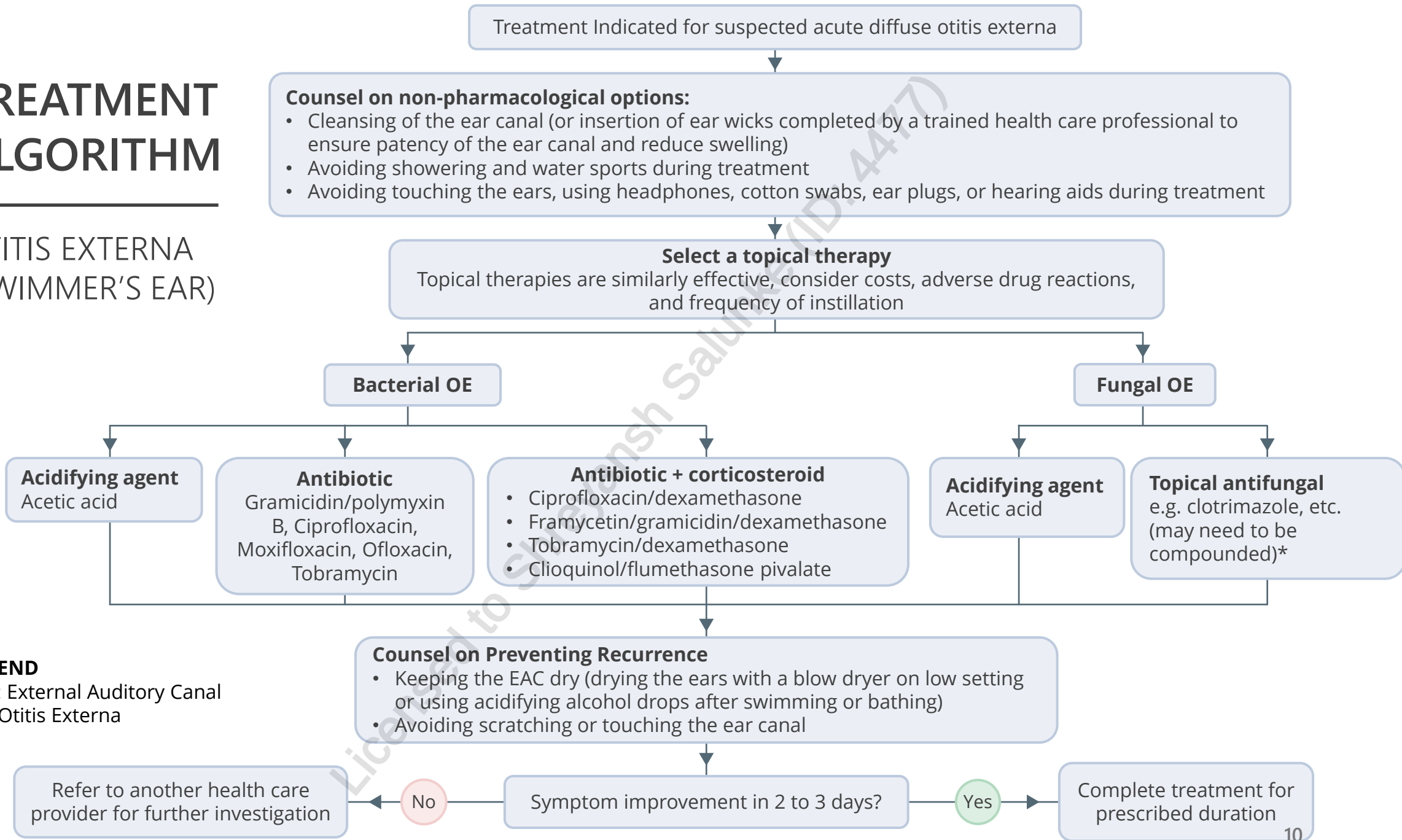
- ✓ Determine the type of OE and possible causes
 - ✓ Swimming, trauma to EAC, comorbid dermatologic conditions on other areas of the body
- ✓ Control otalgia, fever, and discomfort
- ✓ Restore the EAC to normal health (natural acidic pH, adequate cerumen) to prevent future occurrence of infections

TREATMENT ALGORITHM

OTITIS EXTERNA (SWIMMER'S EAR)

LEGEND

EAC: External Auditory Canal
OE: Otitis Externa



NON-PHARMACOLOGICAL MEASURES

- **Cleansing of the ear canal** by removing debris to ensure topical treatment is effective by removing excess cerumen and exudate
 - Cleansing should be done by primary care provider or specialist
 - **Patients should avoid using cotton swabs**
- After swimming or bathing, **dry the external canal** with a blow dryer (low power) or by instilling acidifying or alcohol drops
- **During pharmacological treatment (7-10 days), avoid any water sports**
- Patients believe ear plugs and bathing caps keep the ears dry however there is little evidence to support this; **frequent use of ear plugs can irritate the area and cause infections**

DRUG THERAPY - ACUTE DIFFUSE OE

Topical therapies: acidifying agents, antibiotics (with/without corticosteroids) – see drug charts for information

- **Topical treatments are 1st line**
- Some patients may require systemic agents in severe situations e.g. co-existing otitis media, immunocompromised (beyond the scope of the lecture)
- All topical treatments have been shown to have similar efficacy
- Choosing the correct agent is a **collaborative decision** between HCP-patient: cost, ADR, frequency of instillation etc.
- **Antibiotic drops are available as ophthalmic and otic preparations; otic products are more acidic and may cause burning as a side effect (if patients cannot tolerate, consider trying ophthalmic products)**

DRUG THERAPY - ACUTE DIFFUSE OE (CONTINUED)

Topical therapies: acidifying agents, antibiotics (with/without corticosteroids) – see drug charts for information

- **Aminoglycoside-induced ototoxicity** should be considered with topical treatments (topical fluoroquinolones do NOT cause ototoxicity)
- Ototoxicity has been shown to occur with systemically administered aminoglycosides
- Risk factors for ototoxicity with topical agents: ruptured tympanic membranes (e.g. tympanostomy tubes), **using the product for greater than 7 days, and using the product after otorrhea subsided**
- Note: in fungal OE (less common), cleansing and acidifying agents may suffice, however, some patients may require topical treatments (e.g. clotrimazole) that may be compounded in a specialty pharmacy → **very important to not treat with topical antibiotics because it could promote growth of fungi**

DRUG THERAPY - ACUTE LOCALIZED OE (FURUNCULOSIS)

FYI

Topical mupirocin or fusidic acid for mild symptoms

- Pain subsides when the boil bursts

Incision and drainage for moderate-severe cases

- In some situations, systemic antibiotics targeting *S. aureus* may be required

DRUG THERAPY - CHRONIC OTITIS EXTERNA

Treat the underlying allergic contact dermatitis

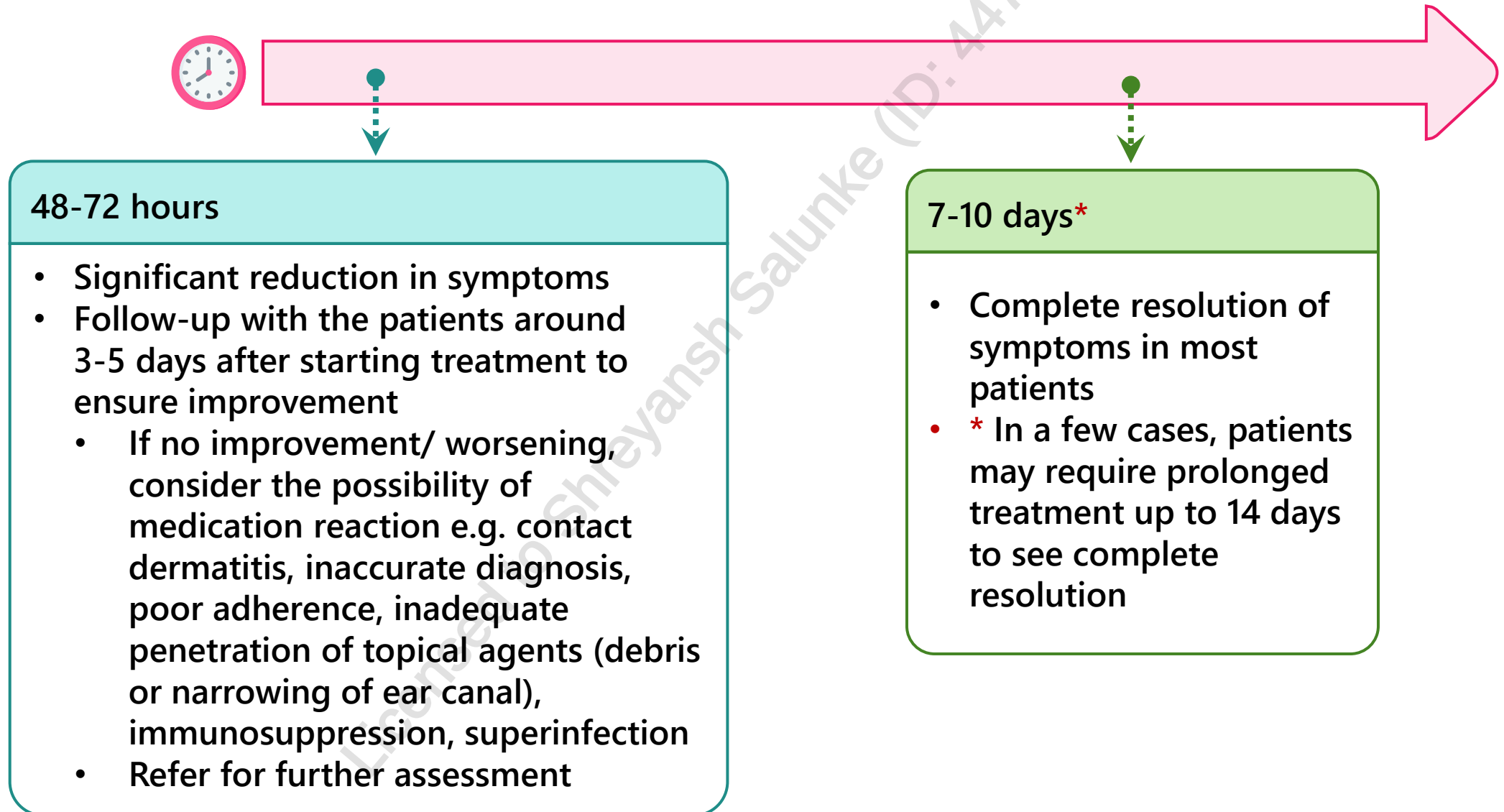
- Commonly due to nonbacterial causes e.g. allergic contact dermatitis
- Benefit from topical steroid cream applied twice daily

DRUG THERAPY - ECZEMATOUS OTITIS EXTERNA

Treat the underlying dermatologic condition (e.g. psoriasis, acne, seborrhea)

- **Allergic** contact dermatitis is common in/on the ears due to grooming products e.g. hair sprays, hair dyes, shampoos
- Hearing aids and ear plugs may cause EAC dermatitis
- Topical agents containing benzalkonium chloride, benzocaine, and propylene glycol are known to cause contact dermatitis
- Tobramycin may also cause allergic contact dermatitis
- Treatment: avoid offending agent and treat symptoms with either topical corticosteroids or calcineurin inhibitors

MONITORING PARAMETERS



DRUG TABLES

Name (topical)	Comments
Acetic acid 2%	• Preventing and treating mild acute OE • Broad-spectrum antibacterial properties • Associated with ototoxicity • No commercial product currently available: prepare by diluting white vinegar with equal parts of either isopropyl alcohol or water
Ciprofloxacin, moxifloxacin, ofloxacin	• Treating acute OE (eyedrops may be used in the ears) • No ototoxicity (preferred choice if tympanic membrane is perforated or can not be visualized) • Gram-negative (e.g. <i>P. aeruginosa</i>) and gram-positive (e.g. <i>S. aureus</i>) activity
Gramicidin/ polymyxin B (Optimycin®)	• Treating acute OE • Associated with ototoxicity • Polymyxin B has gram-negative activity whereas gramicidin has gram-positive activity
Tobramycin	• Treating acute OE (eyedrops may be used in the ears off-label) • Risk of ototoxicity (avoid in patients with perforated tympanic membranes, tympanostomy tubes; avoid using for greater than 7 days) • Gram-negative activity (e.g. <i>Pseudomonas</i>) and gram-positive activity (e.g. <i>S. aureus</i>)

DRUG TABLES

Name (topical)	Comments
Dexamethasone (Maxidex®)	• Treating dermatologic causes of acute OE • Anti-inflammatory activity
Ciprofloxacin/ dexamethasone (Ciprodex®)	• Combination product commercially available for treating acute OE
Clioquinol/ flumethasone pivalate (Locacorten Vioform® eardrops*)	• Combination product commercially available for treating acute OE with fungal etiology • Clioquinol has activity against fungi and gram-positive bacteria whereas the steroid component has anti-inflammatory activity
Framycetin/ gramicidin/ dexamethasone (Sofracort®)	• Combination product commercially available for treating acute OE • Risk of ototoxicity (avoid in patients with perforated tympanic membranes, tympanostomy tubes; avoid using for greater than 7 days) • Framycetin has gram-negative activity (NOT <i>Pseudomonas</i>) and gram-positive activity (e.g. <i>S. aureus</i>); gramicidin has gram-positive activity; dexamethasone has anti-inflammatory activity

DRUG TABLES

Name (topical)	Comments
Tobramycin/ dexamethasone (Tobradex®)	• Treating acute OE (eyedrops may be used in the ears off-label) • Risk of ototoxicity (avoid in patients with perforated tympanic membranes, tympanostomy tubes; avoid using for greater than 7 days) • Gram-negative activity (e.g. <i>Pseudomonas</i>) and gram-positive activity (e.g. <i>S. aureus</i>)
Isopropyl alcohol 95%/ glycerin 5% (Auro-Dri®)	• Used for preventing acute OE (drying agent) • In patients with acute OE, this agent may be painful
Clotrimazole 1%	• For fungal otitis externa • Apply BID for 7 days • Available OTC

ADMINISTRATION

1. Lie on your side so the treated ear is facing up
2. Hold the dropper above the ear canal, but do not let it touch the external auditory canal
3. Straighten out the ear canal
 - Age under 3 years: pull top of the ear down and back
 - Age over 3 years: pull top of the ear up and back
4. Use enough drops to fill the external auditory canal (3 to 4 drops for anyone over 6 months of age)
5. Remain on your side for 3 to 5 minutes so the drops can enter the external auditory canal

* If drops are required for both ears, allow 5 to 10 minutes between administrations

CLINICAL PEARLS

- Both ophthalmic and otic antibiotic drops can be used for treatment
- Ototoxicity risk increases when the tympanic membrane is ruptured and when the treatment is used for more than 7 days
- Topical anesthetics have not been shown to be effective
- Acidifying alcohol drops used to dry the ear canal should not be used during active treatment
- If clinical resolution does not occur in 7 to 10 days, treatment can be extended for up to 2 weeks
- Oral analgesics can be used to treat pain

TAKEAWAY

- Acute diffuse otitis externa is commonly caused by *Pseudomonas aeruginosa* (20-60%) and/or *Staphylococcus aureus* (10-70%), or it may be due to a fungal cause (rare)
- Non-pharmacologic measures include cleaning the ear canal, avoiding the use of cotton swabs or ear plugs, drying the external canal after swimming/bathing, and avoiding water sports for the duration of pharmacologic treatment
- **Topical therapies are first-line** pharmacological agents
 - Bacterial OE therapeutic options: an **acidifying agent** (acetic acid), **antibiotics**, or an **antibiotic + corticosteroid**
 - Fungal OE therapeutic options: an **acidifying agent** (acetic acid) ± a **topical antifungal** (e.g. compounded clotrimazole)
- Antibiotic drops are available as ophthalmic and otic preparations
 - **Otic products are more acidic and may cause burning as a side effect**
- **Aminoglycoside-induced ototoxicity should be considered with topical treatments**
- Topical fluoroquinolones do not cause ototoxicity
- **Fungal OE should not be treated with topical antibiotics as this may promote fungal growth**

CASE

LJ is a 34-year-old woman who presents to the clinic with complaints of severe right-sided otalgia that worsens with chewing, pruritus, reduced hearing, and watery discharge from her right ear. She reports that these symptoms have been ongoing for the past 3 days. LJ tells her physician that she has never experienced this before, and she denies having a fever or any upper respiratory tract infections for the past 6 months. She returned from a holiday last week where she swam every day and used cotton swabs in both ears multiple times a day. An otoscopic exam revealed the external auditory canal to be erythematous, edematous, and tender on tragal manipulation, with the absence of an infected hair follicle. Her past medical history is significant for allergic rhinitis for which she takes loratadine 10 mg PO daily. She is not currently pregnant or breastfeeding. She has an allergy to penicillin which results in widespread urticaria.

1. Which of the following types of otitis externa (OE) is LJ most likely experiencing?
 - a) Eczematous OE
 - b) Acute diffuse fungal OE
 - c) Acute localized bacterial OE
 - d) Acute diffuse bacterial OE



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2. Which of the following pharmacologic agents is most appropriate to recommend for LJ?
- a) Clioquinol/flumethasone pivalate
 - b) Clotrimazole 1% compounded
 - c) Isopropyl alcohol 95%/glycerin 5%
 - d) Ciprofloxacin/dexamethasone



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3. All of the following non-pharmacologic recommendations can be made, EXCEPT:
- a) Avoid the use of headphones during treatment
 - b) Cleansing of the ear canal should be performed daily at home
 - c) Avoid water sports during treatment
 - d) Frequent use of ear plugs can irritate the area and cause infections



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4. What is the appropriate follow-up interval for LJ?
- a) 5 days after symptom onset
 - b) 3 days after starting pharmacotherapy
 - c) 10 days after starting pharmacotherapy
 - d) 14 days after symptom onset



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CHANGE LOG

March 2025

- Slide 3: Added abbreviation for OTC
- Slide 7: Added painless for chronic OE, common pathogen for NOE
- Slide 15: Added "DRUG THERAPY:" on the title for chronic otitis externa for consistency and added examples of management
- Slide 19: Added clotrimazole
- Slide 23: Added references

July 2025

- Content reviewed; minor formatting changes were made
- Updated legend
- Slide 14: Added 'FYI'
- Slides 19 and 20: Formatting changes made

September 2025

- Grammar changes
- Slide 8: Added 'patients with diabetes' to red flags warranting referral
- Slide 23: Takeaway added
- Slides 24, 25, 26, 27: Case study and questions added

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 16: Removed neomycin since not available for human use in Canada
- Slides 17 and 18: Minor updates for clarification